

This note is about cycles in the citation graph; their connection to subsumption and possibly to web pages.

For an acyclic citation graph, I believe that we are all happy that our standard model for credit transfer is reasonable. The problem happens when we introduce cycles where, even though the results for kudos make sense and that we can show that what goes in must come out, we can get large credit for nodes. The simplest example is when we have two documents D_1 and D_2 . D_1 sends all its credit to D_2 and D_2 keeps 10% of the kudos and transfers the remaining 90% to D_1 . Moreover, one unit of credit flows into D_1 .

Following the notation in my “notes” document ($\mathbf{A}$ is the credit transfer matrix; $\mathbf{c}$ $\mathbf{v}$ and $\mathbf{k}$ are vectors which give, at each node, the credit, input and kudos; and $\mathbf{1}$ is a vector of one) we solve

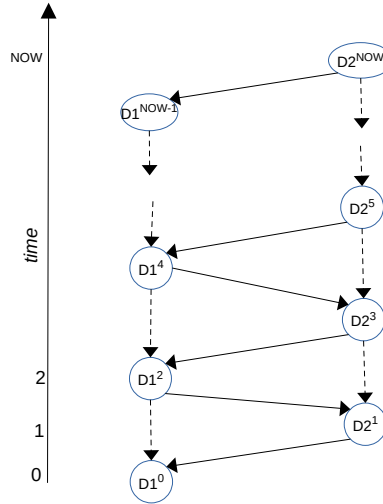
$$\mathbf{c} = \mathbf{A}^\top \mathbf{c} + \mathbf{v} \quad \text{and} \quad \mathbf{k} = \mathbf{c} - (\mathbf{A}\mathbf{1}) \times \mathbf{c} \quad \text{when} \quad \mathbf{A} = \begin{bmatrix} 0 & 1 \\ 0.9 & 0 \end{bmatrix} \quad \text{and} \quad \mathbf{v} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

which gives

$$\mathbf{k} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \mathbf{c} = \begin{bmatrix} 9 \\ 10 \end{bmatrix}$$

Why such a high credit total in each node? In an acyclic graph the credit at any node would surely be less than the total input to all the ancestors of that node.

Maybe to explain this, consider the following acyclic structure representing a simple, but highly artificial set of events.



- At time 0, document D_1^0 is published. There's nothing to cite so it gets to keep all its input as kudos.
- At time 1 D_2^0 is published. It D_1^0 and transfers 90% of its credit to D_1^0 . However D_1^0 does not necessarily receive credit for the citation link
- At time 2, D_1^2 is published, which cites D_2^1 . It decides to transfer all its credit to D_2^1 . But again D_2^1 does not necessarily receive extra credit for the citation link from D_2^1
- ... and so on, alternating until ...
- Time t^{NOW} , the present time. In the picture, NOW is assumed to be odd.

Apart from having the same credit policy I have said nothing about the relationship between $D_1^0, D_1^1, D_1^2 \dots$ and between $D_2^0, D_2^1, D_2^2 \dots$. Nor have I described the credit inputs.

Suppose just one credit is given to D_2^{NOW} then, with $\text{NOW} = 8$, we get

t	D_1^t	D_2^t	k_1^t	k_2^t
8	0.0	1.0	0.0	0.1
7	0.9	0.0	0.0	0.0
6	0.0	0.9	0.0	0.09
5	0.81	0.0	0.0	0.0
4	0.0	0.81	0.0	0.081
3	0.729	0.0	0.0	0.0
2	0.0	0.729	0.0	0.0729
1	0.6561	0.0	0.0	0.0
0	0.0	0.6561	0.0	0.06561

Not so interesting, but now suppose the vertical dotted arrows in the figure above indicate (linear) subsumption; and suppose we $\text{NOW} = 100$. I'm not going to show the whole table, but since we have subsumption it make sense to add up the kudos – and the credit at each node. That is, we add up the columns to get:

$$\mathbf{k} = \begin{bmatrix} 0 \\ 0.995 \end{bmatrix} \quad \mathbf{c} = \begin{bmatrix} 8.95 \\ 9.95 \end{bmatrix}$$

Look familiar? See above. This is an explanation of why we see large credit at nodes.

I'm not sure where this is going, but there are various questions. First about subsumption. If A subsumes B:

- If A (as is often the case) cites B, should this add to the kudos of B – and hence of A (the subsuming document)?
- If C cites both A and B does this count as one or two citations for A?
- If both A and B cite C, does this count as one or two citations for B?

To make credit transfer work, I suspect that the answer to all three questions has to be no.

More questions to follow, but an intriguing question is whether this linear model of evolving subsuming documents also serves as a model for web pages. If so there should be a connection with Page-rank. Maybe it's trivial.